

LAMONI, IA, USA

WMO: 725499

Lat: 40.631N

Lon: 93.901W

Elev: 1135

StdP: 14.10

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 94991

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	1.6	-10.3	3.3	-1.7	-6.4	4.1	3.2	28.2	27.8	26.2	26.7	9.0	310	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	92.4	76.0	89.8	75.3	87.1	74.2	78.7	87.8	77.2	86.3	75.6	83.8	10.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.9	141.3	84.4	74.4	134.3	83.0	72.8	127.0	81.5	43.1	88.4	41.5	86.1	39.9	84.0	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.0	21.9	19.2	-10.9	96.7	5.3	3.0	-14.8	98.9	-17.9	100.7	-20.9	102.3	-24.8	104.5		
			-11.5	81.3	5.2	1.1	-15.2	82.1	-18.2	82.8	-21.1	83.4	-24.9	84.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	24.5	27.6	40.7	52.6	62.2	71.7	75.9	73.7	66.2	53.6	40.7	28.9
	DBStd	20.31	12.17	12.45	11.62	9.52	8.06	5.83	5.58	5.42	7.97	9.48	10.70	11.54
	HDD50	2882	790	631	334	80	7	0	0	0	1	73	311	655
	HDD65	5898	1255	1047	753	381	149	12	2	4	79	369	729	1118
	CDD50	3491	0	4	48	159	386	652	802	736	487	184	32	1
	CDD65	1033	0	0	2	10	64	214	338	275	115	15	0	0
	CDH74	9144	0	0	13	115	489	1799	3292	2331	985	120	0	0
	CDH80	3183	0	0	0	18	101	575	1317	840	314	18	0	0
Wind	WSAvg	9.4	10.5	10.6	10.9	11.1	9.5	8.3	7.3	6.8	7.9	9.4	10.4	10.3
Precipitation	PrecAvg	39.0	1.0	1.5	2.4	4.1	5.7	5.7	5.1	4.3	3.4	3.0	2.0	1.3
	PrecMax	56.6	2.0	4.0	5.3	8.1	12.0	13.3	19.8	9.5	11.0	6.2	5.2	3.9
	PrecMin	24.6	0.2	0.2	0.1	1.3	1.9	1.3	2.0	1.0	0.6	0.4	0.0	0.0
	PrecStd	8.6	0.5	0.9	1.4	1.5	2.5	2.7	4.1	2.6	2.4	1.6	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	65.4	75.5	82.3	87.9	92.4	97.9	96.7	91.1	82.4	72.3	60.9
		MCWB	45.6	51.7	61.2	64.5	72.3	76.2	75.3	73.9	74.0	68.2	59.2	53.8
	2%	DB	50.7	57.4	69.8	77.4	82.5	89.0	93.1	91.1	87.6	77.5	66.0	53.7
		MCWB	43.0	48.7	57.1	61.9	69.1	74.9	76.6	75.1	73.3	65.0	56.7	47.9
	5%	DB	45.5	51.1	64.2	72.8	79.5	86.1	90.3	88.1	83.7	73.0	61.3	48.6
		MCWB	39.6	44.5	54.7	59.6	67.0	73.2	76.4	74.6	70.7	61.9	52.9	43.6
	10%	DB	40.4	45.2	59.2	68.7	76.1	83.2	87.6	84.3	79.8	69.3	56.7	44.8
		MCWB	36.1	39.6	51.3	57.7	65.2	71.8	75.2	73.1	68.3	60.4	49.7	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	56.0	63.5	67.4	74.9	79.1	81.2	80.5	76.9	71.8	61.7	56.0
		MCDB	53.2	60.3	72.1	78.6	85.2	88.9	90.2	88.5	87.1	79.2	67.8	59.5
	2%	WB	43.6	49.5	60.1	64.5	71.6	76.7	79.2	78.1	74.8	68.1	58.1	49.6
		MCDB	49.1	57.6	66.0	74.0	80.2	85.7	88.6	86.6	84.0	73.5	64.1	51.6
	5%	WB	40.2	44.5	56.7	61.7	69.1	74.9	77.8	76.4	72.9	64.5	54.5	44.2
		MCDB	45.8	50.4	63.5	70.7	77.3	83.5	87.1	84.6	81.1	69.9	60.0	48.2
	10%	WB	36.3	39.5	51.3	58.9	66.9	73.3	76.3	74.6	70.8	61.0	50.6	40.1
		MCDB	40.2	44.7	58.8	67.0	74.2	81.6	84.6	82.3	77.6	67.8	56.1	44.4
Mean Daily Temperature Range	5% DB	MDBR	17.0	17.8	19.0	20.2	18.6	18.3	18.6	18.6	20.3	19.5	18.0	16.1
		MCDBR	23.2	25.3	25.2	25.9	21.5	20.5	21.2	22.2	22.4	23.6	23.7	21.5
		MCWBR	16.8	17.7	14.8	15.1	11.7	10.2	8.8	9.8	10.1	13.1	15.3	16.4
	5% WB	MCDBR	21.9	23.6	21.8	22.9	19.1	18.8	18.6	19.0	19.5	19.0	21.2	19.6
		MCWBR	16.5	17.9	13.6	14.9	11.4	10.6	9.6	10.6	10.1	13.2	15.4	16.5
Clear-Sky Solar Irradiance	taub	0.288	0.316	0.331	0.386	0.411	0.419	0.429	0.423	0.379	0.335	0.296	0.284	
	taud	2.438	2.361	2.402	2.264	2.260	2.279	2.247	2.285	2.357	2.475	2.522	2.471	
	Ebn at Noon	279	284	292	281	275	272	268	266	272	274	274	271	
	Edh at Noon	25	31	34	42	43	42	43	40	35	28	23	22	
All-Sky Solar Radiation	RadAvg	633	928	1203	1530	1804	2001	2012	1754	1493	1014	698	543	
	RadStd	60	68	100	102	148	180	149	100	100	115	59	56	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page